

1 Introduction

In this report we go over the work that we have done since (and before) receiving the car until now.

Overall our activities this sprint was focused on researching and planning what would be our strategy to tackle the challenge, understanding the brain code and trying to add some autonomous features to it. In general our activity during this phase was low because all of the team members had exams during January. Once this exam session is done we are all completely free to work full time on the competition.

2 Planned activities

We decided to split our efforts in research and preparation. We planned for 4 main objectives for this phase.

1-Neural Networks : research what kind of NN will we use and which hardware do we need to train and run it, Michele and Davide were responsible for this part.

2-Getting the simulator to run and beginning to test code on it, Panagoda was responsible for this part.

3-The embedded system , looking at which sensors and hardware will we need, Banaan was responsible for this part.

4-Implementing basic autonomous functionality of obstacle avoiding and testing and calibrating it, Ahmed was responsible for this part.

3 Status of planned activities

1-Neural Networks: A CNN Network will be used to train on detecting and classifying multiple objects. More details still under review/research.

2-Simulator : The simulator is working but we are facing challenges in getting the camera to work inside it, currently debugging.

3-Embedded system: Postponed until Neural Network capabilities and computational load are better understood

4-Autonomous functionality: The first autonomous function we decided to implement was obstacle avoidance. We started to implement an obstacle avoiding algorithm based on a set of thresholds taking as input the output of the canny edge detector algorithm.

In particular, we obtain the roi (region of interest) from the camera sensor as the lower region of what camera detects. We then use Canny edge detector on the image and we find the contours on its output. We then filter the contours and ignore small ones, and then categorize them into left, right and center. Based on this information the car decides what is the steering angle and if the number of contours is higher than a threshold (obstacle is close) we command the velocity to a low value to be able to steer away before hitting the obstacle.

4 General status of the project

The car can be controlled manually and the calibration is done. The car can also be controlled autonomously via the camera to detect and avoid obstacles.



5 Upcoming activities

All of the planned activities need further work to be completed. The next step would be a lane keeping algorithm and a state machine that determines the high level logic of the car and which actions it needs to take

github: <https://github.com/UniPD-DriveOps-BFMC/Brain>

The branch that has the latest code used for obstacle avoiding is pi_testing